

Continuum multiband exact diagonalization of twisted MoTe₂

An ED benchmark for CDW and fractional Chern-insulator neural states

Local numerical benchmark for the Physics-for-AI/QNS programme

22 August 2026

Abstract

We establish a symmetry-resolved exact-diagonalization (ED) benchmark and a continuous-coordinate neural-Bloch experiment for twisted MoTe₂. The continuum calculation reproduces the one-five-band reference energies, while one-band 27-cell ED distinguishes the $\nu = 1/3$ charge-density-wave tower from the approximately three-fold Γ manifold at $\nu = 2/3$. A tracked three-state calculation gives Chern numbers near zero and two, but complete all-sector spectral isolation, large-gauge sewing and twist-grid convergence remain required before assigning a quantized Hall response.

The variational state is one generalized determinant of neural orbitals in the full plane-wave continuum basis and follows the Luo-Fu Appendix-B construction. Four 4,128-walker, 120-update runs compare full momentum mixing with an exact- Γ no- M construction. Ordinary CG is stopped between 20 and 60 iterations using an explicitly recomputed true residual; unstable directions are rejected and trigger a temporary larger damping and smaller learning rate. Fresh-walker validation selects no M at both fillings. Independent 4,128-walker, 300-burn-sweep precise validation gives $E/N_e = -36.78317(1019)$ and $-52.39973(1673)$ meV at $\nu = 1/3$ and $2/3$, remaining 0.6101 and 0.3258 meV per particle above five-band ED. Full-ratio five-band-projected 1-RDM estimates give bare-band-1 weights of 96.63(94)% and 89.12(86)%, while direct two-body structure factors preserve the CDW/FCI contrast. The current single-seed finite optimization therefore shows no variational energy advantage and does not support a converged natural-orbital or topological interpretation.

Part I

Exact-diagonalization benchmark

1 Motivation

Fractional Chern insulators (FCIs) in moiré semiconductors require simultaneous control of strong Coulomb correlations, band mixing and nonuniform Bloch-state quan-

tum geometry. A quantum neural state (QNS) can reach larger systems and avoid a rigid few-band truncation, but a lower variational energy is not sufficient evidence of the correct phase. The reference calculation must also reproduce symmetry quantum numbers, the torus ground-state counting, momentum occupation and density correlations.

The present benchmark therefore separates two questions. First, the 3×3 multiband calculation tests the continuum Hamiltonian and the variational convergence from one to five retained bands, directly against Fig. 2 of Ref. [1]. Second, the one-band 27-site calculation repeats the phase diagnostics of Supplementary Fig. 5: a Γ, K, K' CDW tower at $\nu = 1/3$ versus an approximately three-fold Γ manifold at $\nu = 2/3$ [1]. The latter momentum counting is supplemented below by flux spectral flow and a gauge-invariant many-body Chern number.

2 Continuum model and quantum geometry

For a fixed spin/valley sector, the single-hole Hamiltonian is

$$H_0 = \begin{pmatrix} \frac{\hbar^2(-i\nabla - \kappa_+)^2}{2m^*} + V_1(\mathbf{r}) & t(\mathbf{r}) \\ t^*(\mathbf{r}) & \frac{\hbar^2(-i\nabla - \kappa_-)^2}{2m^*} + V_2(\mathbf{r}) \end{pmatrix}. \quad (1)$$

The lowest moiré harmonics and plane-wave convention follow Refs. [5, 1]. We use

$$(V, w, \phi, m^*) = (11.2 \text{ meV}, 13.3 \text{ meV}, -91^\circ, 0.62m_e), \quad (2)$$

monolayer lattice constant $a_0 = 0.352$ nm, and eight radial reciprocal shells. The resulting Bloch Hamiltonian has 61 plane waves per layer. Figure 1 compares the first ten moiré bands at $\theta = 2.6^\circ$ and 3.0° .

For a nondegenerate band n , the quantum geometric tensor is evaluated from velocity matrix elements,

$$Q_{\alpha\beta}^{(n)}(\mathbf{k}) = \sum_{m \neq n} \frac{\langle n|v_\alpha|m\rangle\langle m|v_\beta|n\rangle}{[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2}, \quad (3)$$

with $g_{\alpha\beta} = \text{Re } Q_{\alpha\beta}$ and $\Omega = -2 \text{Im } Q_{xy}$. Bands 1–3 are calculated on a 49-point linear grid in each reciprocal direction, restricted to the hexagonal Brillouin zone, and smoothly triangulated only for visualization. The shared colorbars retain physical one-dimensional numerical ticks. Values outside the 1.5–98.5 percentile interval are clipped in the rendering, not in the saved data.

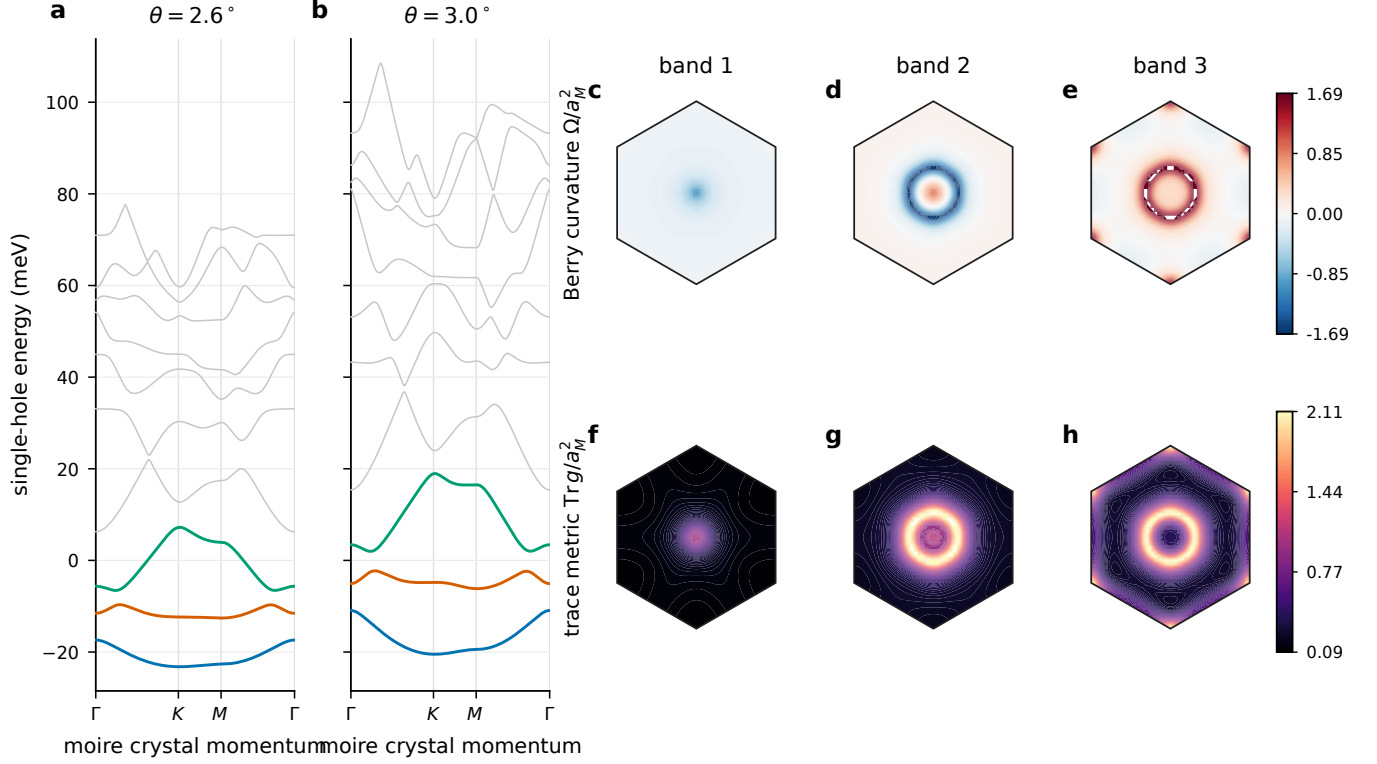


Figure 1: Continuum bands and quantum geometry. (a,b) The first ten moiré bands along Γ - K - M - Γ at $\theta = 2.6^\circ$ and 3.0° ; bands 1–3 are colored blue, orange and green, respectively. (c–e) Berry curvature and (f–h) trace quantum metric for bands 1–3 at $\theta = 2.6^\circ$ across the hexagonal Brillouin zone. Both rows use a common physical color scale with explicit numeric ticks.

3 Projected interaction and ED symmetries

The periodically continued Coulomb interaction is projected into N_b Bloch bands,

$$H = \sum_{n\mathbf{k}} \varepsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}} + \sum_{ab,cd} V_{cd,ab} c_c^\dagger c_d^\dagger c_b c_a, \quad (4)$$

where $V(\mathbf{q}) = 2\pi e^2/(\epsilon|\mathbf{q}|)$ and the uniform $\mathbf{q} = 0$ component is removed by the neutralizing background. Eight radial reciprocal shells are retained in the form-factor sum.

Charge- $U(1)$ is enforced by fixed hole number, while spin- $U(1)$ is enforced by restricting to the fully polarized sector. Translation symmetry first decomposes the Hilbert space into many-body momentum blocks $\mathcal{H}_K$. At generic momentum, a rotation maps $\mathcal{H}_K$ to the distinct block $\mathcal{H}_{C_n K}$, so point-group symmetry is used only to copy spectra within a momentum orbit. A further C_3/C_6 character decomposition is available only when $C_n K = K + G$, i.e. when the rotation belongs to the little group of K .

The phase-diagnostic cluster has real-space supercell matrix

$$S_{27} = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad \det S_{27} = 27, \quad (5)$$

and is therefore the $3\sqrt{3} \times 3\sqrt{3}$ C6 torus rather than a rectangular 3×9 mesh. Its momentum group is written

as $\mathbb{Z}_3 \times \mathbb{Z}_9$, with plot index

$$I_k = k_1 + N_1 k_2, \quad N_1 = 3, \quad I_k = 0, \dots, 26. \quad (6)$$

In this convention Γ, K', K occur at $I_k = 0, 9, 18$. The one-band Hilbert space contains $\binom{27}{9} = 4,686,825$ states at $\nu = 1/3$. At $\nu = 2/3$, an exactly verified particle-hole transformation maps 18 particles to 9 holes without changing the spectrum.

4 Multiband benchmark on the 3×3 cluster

Figure 2 summarizes the 3×3 calculation at $\theta = 3^\circ$, $\epsilon = 5$. The ground-state energy decreases monotonically as the retained band space grows. The values in Table 1 reproduce the available $\nu = 2/3$ points in Fig. 2 of Ref. [1] within 0.036 meV per hole. Remote-band mixing is visibly stronger at $\nu = 2/3$.

Table 1: Ground-state energy per hole in meV at $\theta = 3^\circ$, $\epsilon = 5$ on the 3×3 cluster.

N_b	$\nu = 1/3$	$\nu = 2/3$	paper Fig. 2
1	-36.3296	-47.1561	-47.15
2	-37.1158	-50.0774	-50.06
3	-37.3518	-51.8605	-51.83
4	-37.3878	-52.5379	—
5	-37.3932	-52.7255	-52.69

The middle column of Fig. 2 shows the lowest energy in every momentum sector for one-, three- and five-band ED. Energies are plotted as $E_K - E_{\text{GS}}$ in meV, without division by particle number. The right column reports the additive band-resolved weight

$$w_b = \frac{\langle N_b \rangle}{N_h}, \quad \sum_{b=1}^{N_b} w_b = 1. \quad (7)$$

For five bands, the weights are

$$\nu = 1/3 : (0.96469, 0.02953, 0.00523, 0.00049, 0.00005), \quad (8)$$

$$\nu = 2/3 : (0.88952, 0.07591, 0.02535, 0.00763, 0.00159). \quad (9)$$

These are normalized band populations, not overlaps between separately optimized truncated Hilbert spaces.

5 Reproduction of Supplementary Fig. 5

Figure 3 repeats the one-band 27-site benchmark at $\theta = 2.6^\circ$, $\epsilon = 5$, and adds the ground-manifold averaged $n(\mathbf{k})$. At $\nu = 1/3$, the three global low states occur at Γ, K, K' with

$$(E_\Gamma, E_K, E_{K'}) - E_{\text{GS}} = (0, 0.573, 0.573) \text{ meV}. \quad (10)$$

The next state is 1.476 meV above the top of this manifold. The K, K' states fold to Γ in the Brillouin zone of the tripled CDW unit cell, matching the interpretation of Ref. [1].

At $\nu = 2/3$, the three lowest states all lie in the Γ sector,

$$(E_{\Gamma,0}, E_{\Gamma,1}, E_{\Gamma,2}) - E_{\text{GS}} = (0, 0.108, 0.298) \text{ meV}. \quad (11)$$

Thus the finite-size threefold splitting is small but explicitly nonzero. The gap between the highest manifold state and the next global level is 3.482 meV. This reproduces the torus counting required for the $\nu = 2/3$ FCI/FQAH state.

The absolute energy origins used to construct the two spectra in Fig. 3 are listed explicitly in Table 2. They are the global minima over all 27 translation sectors; the plotted ordinate is obtained by subtracting the corresponding total energy in the fourth column, without dividing the excitation energy by particle number.

Table 2: Absolute one-band, 27-cell ground-state energies used as the zero of the Fig. 3 spectra.

ν	N_h	three-state manifold	E_{GS} (meV)	E_{GS}/N_h (meV)
1/3	9	Γ, K, K'	-428.359163	-47.595463
2/3	18	$\Gamma_0, \Gamma_1, \Gamma_2$	-1034.203144	-57.455730

For each filling, $n(\mathbf{k})$ and the projected static structure factor are averaged over the three-state ground manifold,

$$n(\mathbf{k}) = \langle c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \rangle, \quad (12)$$

$$S(\mathbf{q}) = \frac{1}{N_h} \langle \rho(-\mathbf{q}) \rho(\mathbf{q}) \rangle, \quad \mathbf{q} \neq 0. \quad (13)$$

The connected value $S(0)$ is set to zero. The $n(\mathbf{k})$ sums are exactly 9 and 18. For Fig. 3, the discrete $n(\mathbf{k})$ markers are placed over an interpolated, weak band-1 Berry-curvature background. The $S(\mathbf{q})$ markers in both rows are placed over the same interpolated band-1 trace-condition deviation $\text{Tr } g_1(\mathbf{q}) - |\Omega_1(\mathbf{q})|$. The background palettes match Fig. 1, while the observable markers use brighter independent color scales.

At $\nu = 1/3$, $S(K) = S(K') = 1.057$, producing the Bragg peaks characteristic of CDW order. At $\nu = 2/3$, the maximum at the same momenta is only 0.197 and is comparable to the surrounding finite- q response; no CDW Bragg peak is present. Together with the three Γ states, this is consistent with the FQAH liquid signature reported in Supplementary Fig. 5.

6 Twisted-boundary spectra and many-body Berry curvature

Let $\mathbf{a}_1, \mathbf{a}_2$ be the primitive moiré real-space vectors and let $\mathbf{b}_a \cdot \mathbf{a}_b = 2\pi\delta_{ab}$. The C_6 -compatible 27-cell ED torus is generated by the columns of

$$S = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad |\det S| = 27, \quad (14)$$

$$\mathbf{L}_1 = 3\mathbf{a}_1 + 3\mathbf{a}_2,$$

$$\mathbf{L}_2 = -3\mathbf{a}_1 + 6\mathbf{a}_2.$$

For every particle i , the many-body twisted boundary condition is

$$\Psi(\dots, \mathbf{r}_i + \mathbf{L}_a, \dots) = e^{i\phi_a} \Psi(\dots, \mathbf{r}_i, \dots), \quad \phi_a \in [0, 2\pi]. \quad (15)$$

Writing a one-particle momentum as $\mathbf{k} = B\mathbf{f}$, with $B = (\mathbf{b}_1, \mathbf{b}_2)$, Eq. (15) gives $S^T \mathbf{f} = \mathbf{m} + \phi/(2\pi)$. Hence the 27 allowed momenta are

$$\mathbf{f}_{\mathbf{m}}(\phi) = S^{-T} \left(\mathbf{m} + \frac{\phi}{2\pi} \right), \quad \mathbf{m} \in \mathbb{Z}^2 / S^T \mathbb{Z}^2, \quad (16)$$

$$\mathbf{k}_{\mathbf{m}}(\phi) = \mathbf{k}_{\mathbf{m}}(0) + \delta\mathbf{k}, \quad \delta\mathbf{k} = BS^{-T} \frac{\phi}{2\pi}. \quad (17)$$

The code enumerates this quotient group with exact integer residues, so precisely 27 inequivalent momenta are

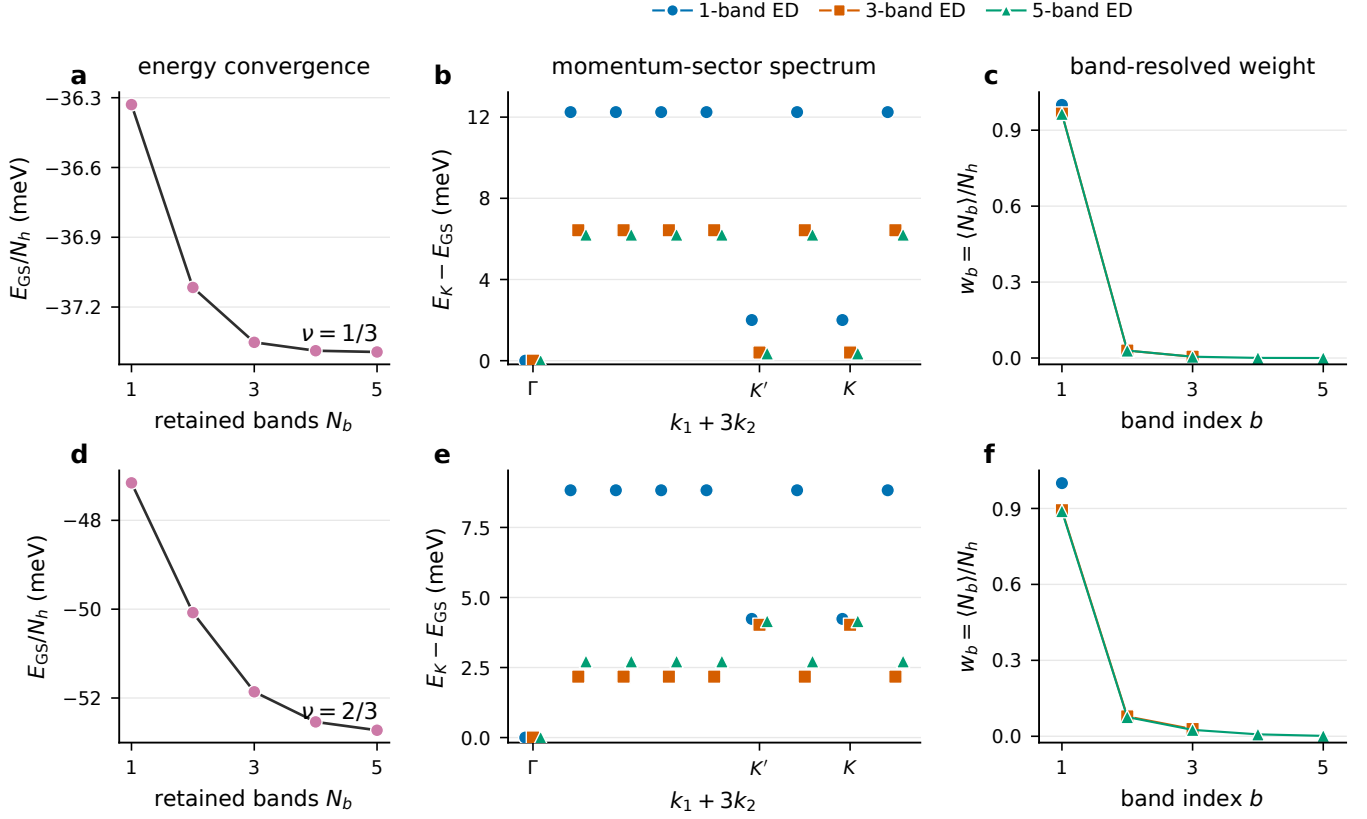


Figure 2: Multiband ED on the 3×3 cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Ground-state energy per hole versus retained bands. (b,e) Momentum-sector minima for one-, three- and five-band ED, plotted as $E_K - E_{GS}$ in meV. (c,f) Normalized band-resolved ground-state weights. Charge- $U(1)$, spin- $U(1)$ and translations are exact; C_3/C_6 equivalence accelerates the largest five-band sectors.

retained and every two-body momentum addition is performed modulo $S^T \mathbb{Z}^2$. The torus is therefore implemented periodically without rounding Cartesian momenta.

At every (ϕ_1, ϕ_2) , the continuum Hamiltonian is solved anew at all $\mathbf{k}_m(\phi)$, after which the one-band Coulomb form factors and projected interaction are rebuilt from the shifted Bloch spinors. A projected scattering term is retained only when $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4$ in the finite translation group (equivalently up to moiré and supercell reciprocal vectors). Charge- $U(1)$ and translations remain exact. A generic twist breaks C_3/C_6 , so rotational orbit copying is not used away from little-group-invariant twists. At $\nu = 2/3$, the 18-electron Hamiltonian is evaluated exactly through its nine-hole particle-hole transform, including its constant and induced one-body terms; this leaves the boundary condition and spectrum unchanged.

Because the three lowest states are quasi-degenerate, an individual-eigenvector Berry phase is gauge dependent. We instead use the determinant link of the full three-state

subspace,

$$M_\mu^{ab}(\phi) = \langle \Psi_a(\phi) | \Psi_b(\phi + \Delta_\mu) \rangle, \quad (18)$$

$$U_\mu(\phi) = \frac{\det M_\mu(\phi)}{|\det M_\mu(\phi)|}, \quad (19)$$

$$\Phi_{12}(\phi) = \arg[U_1(\phi)U_2(\phi + \Delta_1)U_1^*(\phi + \Delta_2)U_2^*(\phi)]. \quad (20)$$

The overlap in M_μ is not a bare dot product of ED coefficient vectors, because the one-body Bloch basis itself changes with twist. For an electron configuration I , its coefficient overlap is weighted by

$$w_I = \prod_{p \in I} \langle u_p(\phi) | u_p(\phi + \Delta_\mu) \rangle. \quad (21)$$

For the nine-hole representation, the code uses the exactly equivalent electron complement,

$$w_H = \frac{\prod_{p=1}^{27} \langle u_p(\phi) | u_p(\phi + \Delta_\mu) \rangle}{\prod_{h \in H} \langle u_h(\phi) | u_h(\phi + \Delta_\mu) \rangle}. \quad (22)$$

The non-Abelian FHS curvature and discrete Chern num-

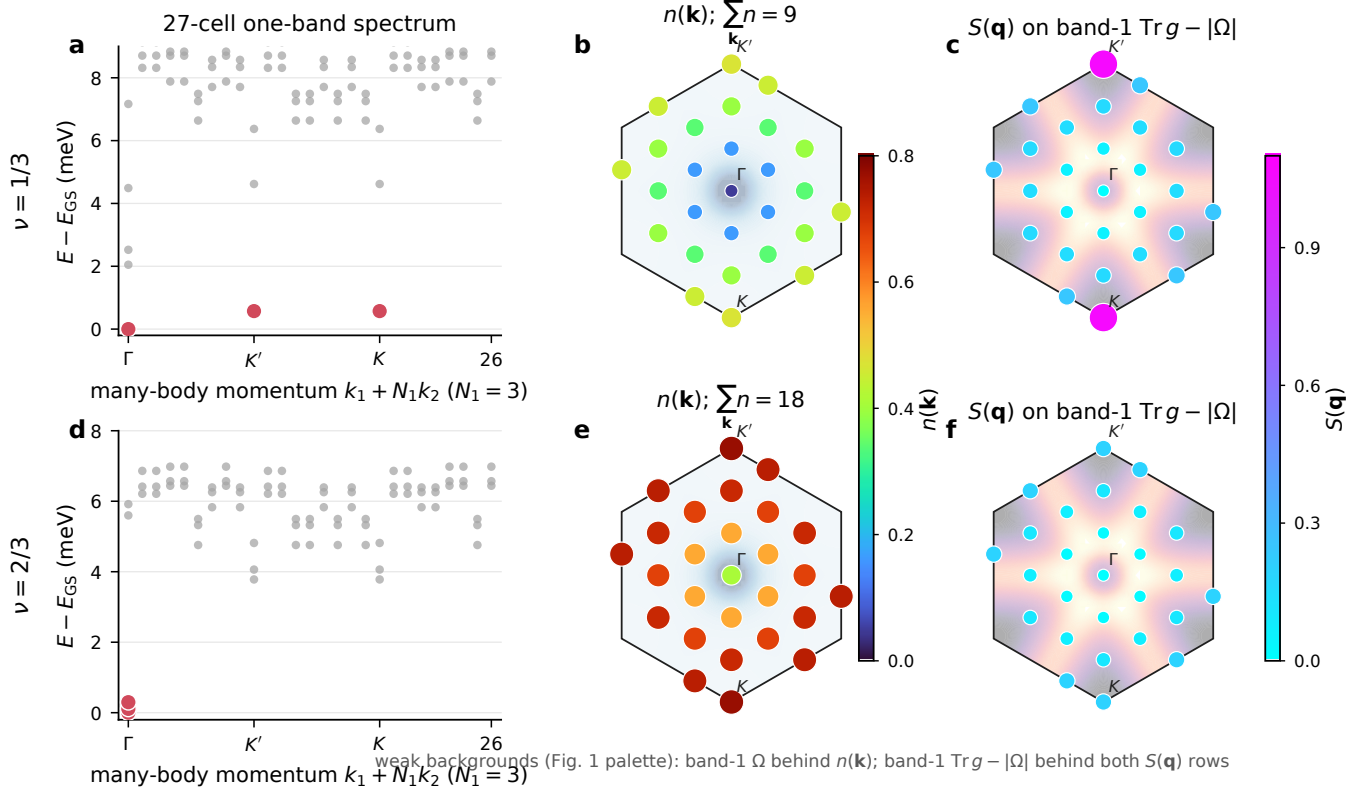


Figure 3: 27-site one-band reproduction of Supplementary Fig. 5. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Momentum-resolved spectra, plotted as $E - E_{\text{GS}}$ in meV. Red markers denote the CDW or FCI three-state manifold. (b,e) Ground-manifold averaged $n(\mathbf{k})$ over weak, interpolated band-1 Berry curvature. (c,f) Projected $S(\mathbf{q})$ over the identical weak, interpolated band-1 $\text{Tr } g_1 - |\Omega_1|$ background. Bright marker colorbars encode the observables independently. The sharp zone-corner peaks at $\nu = 1/3$ and their absence at $\nu = 2/3$ distinguish the CDW and FCI responses.

ber are

$$\begin{aligned} \mathcal{F}_{\phi_1 \phi_2} &= \frac{\Phi_{12}}{(\Delta\phi)^2}, \\ C_{\text{bundle}} &= \frac{1}{2\pi} \sum_{i,j} \Phi_{12}(\phi_{ij}), \\ \Delta\phi &= \frac{2\pi}{6}. \end{aligned} \quad (23)$$

We solve and cache the 7×7 grid endpoints as well as the 6×6 plaquettes. Links crossing a 2π boundary are consequently evaluated from endpoint Bloch overlaps, rather than by identifying raw ED eigenvectors. A separate analytic large-gauge sewing-matrix test is nevertheless still needed for a formal convergence claim. This single grid is a discrete diagnostic, not a twist-grid convergence study.

Figure 4 shows the two axial flux spectra and the curvature map. For $\nu = 1/3$, the bundle contains the lowest states from Γ, K, K' . Its Chern number is

$$C_{\text{bundle}}^{1/3} = 9.3 \times 10^{-7} \simeq 0, \quad (24)$$

and the smallest sampled separation between the three-state manifold and the nearest state in these three sectors is 1.476 meV. For $\nu = 2/3$, the three states all belong to Γ and

$$C_{\text{tracked}}^{2/3} = 1.99999975 \simeq 2. \quad (25)$$

If this tracked subspace is subsequently shown to be the globally isolated ground-state bundle, the conditional Hall response is $\sigma_{xy} = C_{\text{tracked}} e^2 / (3h) = 2e^2 / (3h)$. The minimum same- Γ separation is 5.144 meV, the minimum link singular value is 0.958 at $\nu = 2/3$, and the minimum tracked-sector overlap is 0.964 at $\nu = 1/3$. These checks show that the sampled links are nonsingular, but they do not prove isolation from the complete spectrum. Missing checks are the lowest relevant levels in all 27 translation sectors at every twist, the global $\min_{\phi}(E_4 - E_3)$, an explicit analytic 2π large-gauge sewing test, and $6 \times 6 \rightarrow 9 \times 9 \rightarrow 12 \times 12$ grid convergence.

7 ED benchmark protocol for QNS

The validated workflow is:

1. solve the plane-wave continuum Hamiltonian and verify shell convergence;
2. calculate band dispersion, Berry curvature and quantum metric on the same Bloch convention;
3. project the neutral-background Coulomb interaction with reciprocal-vector form factors;

27-cell one-band ED under twisted boundaries ($\theta = 2.6^\circ$, $\varepsilon = 5$)

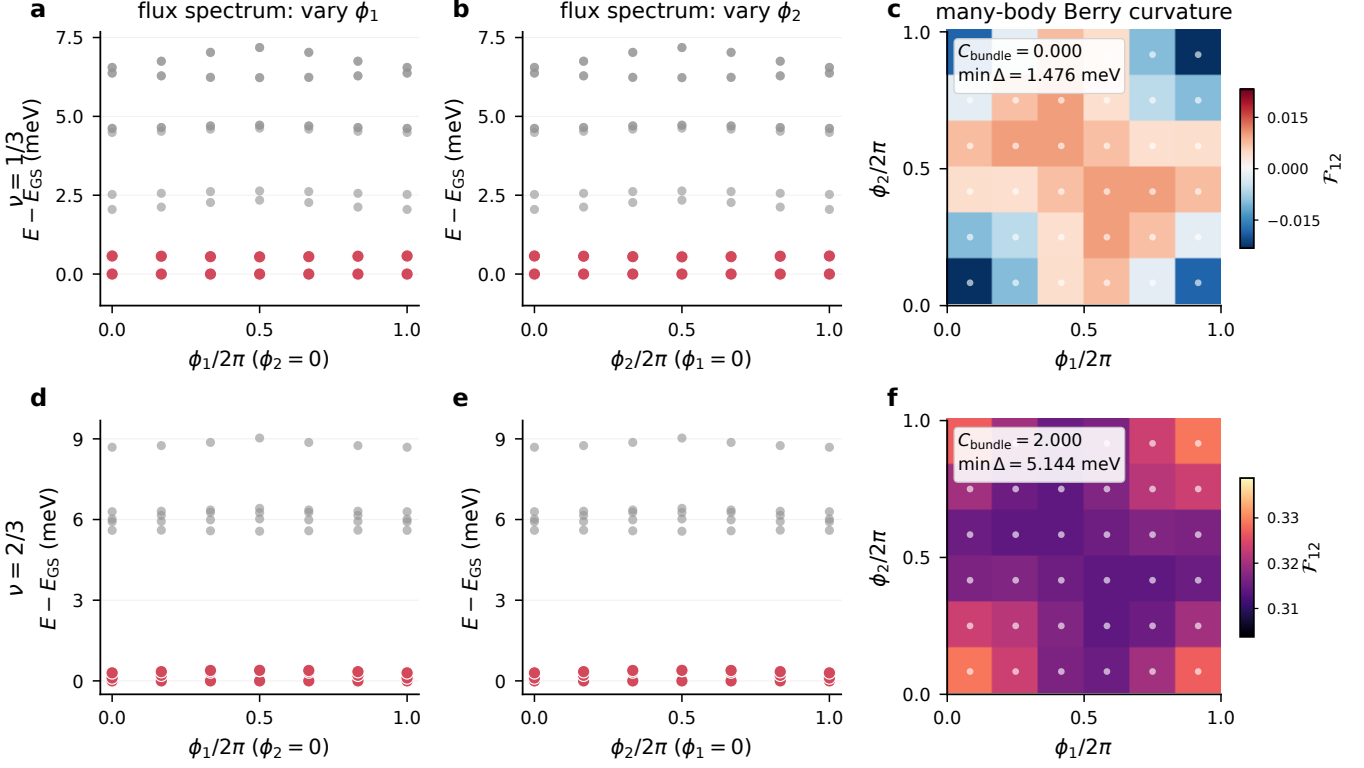


Figure 4: Twisted-boundary ED and many-body topology on the 27-cell one-band cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Spectral flow versus ϕ_1 at $\phi_2 = 0$. (b,e) Spectral flow versus ϕ_2 at $\phi_1 = 0$. At every twist, the lowest three energies are red scatter points. Gray points show five additional $\nu = 2/3$ levels and nine additional $\nu = 1/3$ levels sampled from the three CDW sectors. (c,f) Non-Abelian curvature $\mathcal{F}_{\phi_1\phi_2}$ on the original 6×6 primitive twist grid; white dots mark the FHS plaquette centers. The raw plaquette-flux sums give Chern number zero for the CDW and two for the tracked FCI manifold.

4. fix charge- $U(1)$ and spin- $U(1)$, then block by translation; use point-group orbit equivalence at generic K , and split by C_3/C_6 character only when the rotation belongs to the little group of K ;
5. use sparse Hermitian Lanczos independently in each required sector;
6. compare energies, sector counting, normalized band weights, $n(\mathbf{k})$ and $S(\mathbf{q})$;
7. insert boundary flux, track the isolated ground manifold and integrate its non-Abelian Berry curvature;
8. only then benchmark the QNS energy, variance and state overlaps against ED.

Numerical regression checks support every stage. The general C_6 -supercell interaction agrees with the earlier rectangular 3×3 implementation to relative error 3.3×10^{-16} . Direct electron and particle-hole spectra agree sector by sector to 3.4×10^{-13} meV. The finite plane-wave cutoff produces a maximum one-body C_6 mismatch of 8.0×10^{-6} meV, and the pre-symmetrization interaction Hermiticity error is 7.8×10^{-9} . The current complete 51-test regression suite passes; the topology outputs ad-

ditionally record link conditioning, Lanczos residuals and interaction Hermiticity at every twist.

Part II

Neural many-body Bloch wavefunctions

8 Motivation and scope

The second calculation tests the neural many-body Bloch construction of Ref. [1] directly in the continuum model benchmarked in Part I. The purpose is not to fit coefficients in the one-band ED Hilbert space. Instead, a graph network dresses continuum Bloch orbitals in coordinate space and the resulting generalized Slater determinant is optimized against the complete interacting continuum Hamiltonian. This can generate remote-band content even though the fixed one-body input is the mean-field first band.

This revision follows Appendix B, Eqs. (16)–(23), of Ref. [1]. In particular, the orbital transformation is the

direct complex output

$$J_{\mathbf{k}i}(X) = \text{MLP}([\mathbf{k}, V_i^t]) \in \mathbb{C}, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i} \varphi_{\mathbf{k}i}. \quad (26)$$

There is no constant-one skip, residual parameterization, or factor 0.2; the implemented transformation is the direct $J_{\mathbf{k}i}$ above. The current source additionally implements three audit corrections: full- M uses an unbiased complex random initialization; the node aggregation is the literal $\sum_j m_{ij}$ in Eq. (19), including $j = i$; and the random auxiliary features h_i^0, h_{ij}^0 are explicitly retained in the trainable parameter tree. Earlier one-hot- M , no-self-message checkpoints are excluded from this Part. Figures 6 and 7 and all result tables below are regenerated from the four controlled 4,128-walker, 120-update adaptive-CG runs and their independent random-start validations.

9 Configuration, fixed Bloch basis, and conventions

A walker is a first-quantized configuration

$$X = \{(\mathbf{r}_i, l_i)\}_{i=1}^{N_e}, \quad \mathbf{r}_i \in [0, 1]^2, \quad l_i^{\text{code}} \in \{0, 1\}. \quad (27)$$

The fractional coordinate is converted to a Cartesian coordinate on the 3×3 torus. The layer input used by the network is

$$s_i = 2l_i^{\text{code}} - 1 \in \{-1, +1\}, \quad (28)$$

which is the $l_i = \pm 1$ convention of Ref. [1]. Particle labels are only row labels of the determinant and graph; no ordered occupation string is used.

The nine mesh momenta and plane-wave orbitals are

$$\mathbf{k}_{m_1 m_2} = \frac{m_1}{3} \mathbf{b}_1 + \frac{m_2}{3} \mathbf{b}_2, \quad m_1, m_2 = 0, 1, 2, \quad (29)$$

$$\phi_{\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{1\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{r}}. \quad (30)$$

The reciprocal set contains the same 61 $\mathbf{G}$ vectors per layer used for the single-particle bands in Fig. 1. The coefficients $u_{1\mathbf{k}\mathbf{G}}^l$ are obtained once by diagonalizing the continuum mean-field Hamiltonian. In JAX they are constants, and in PyTorch they are registered buffers. They are absent from the trainable parameter tree. Thus every run uses band 1 as its mean-field orbital source; neither ED amplitudes nor trainable plane-wave coefficients are used.

10 Appendix-B graph network

10.1 Initial physical and auxiliary features

The three reciprocal harmonics used at each vertex are $\mathbf{g}_a \in \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_1 + \mathbf{b}_2\}$. The immutable node feature is

$$v_i^0 = [\sin(\mathbf{g}_a \cdot \mathbf{r}_i), \cos(\mathbf{g}_a \cdot \mathbf{r}_i), s_i] \in \mathbb{R}^7. \quad (31)$$

For every ordered pair (i, j) , the immutable edge feature is

$$e_{ij}^0 = [\sin(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), \cos(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), s_i] \in \mathbb{R}^{19}, \quad (32)$$

where all nine mesh momenta are included. In particular, the edge layer channel is the source layer s_i , exactly as in Eq. (16) of Ref. [1]; no additional layer channels are appended.

The complete initial features are

$$V_i^0 = (v_i^0, h_i^0) \in \mathbb{R}^{32}, \quad E_{ij}^0 = (e_{ij}^0, h_{ij}^0) \in \mathbb{R}^{32}. \quad (33)$$

The code therefore uses 25 auxiliary node components and 13 auxiliary edge components. Appendix B explicitly calls h_i^0 and h_{ij}^0 “randomly initialized parameters.” They are therefore leaves of the trainable JAX pytree (and `nn.Parameter` objects in PyTorch), so natural-gradient updates act on them together with the other network parameters. The present code uses one random vector shared over particle labels and one shared over ordered pairs; sharing preserves permutation equivariance. Each component is initialized from $\mathcal{N}(0, 1/32)$. The paper does not specify this distribution or sharing rule, so both remain explicit local conventions. There is no learned input embedding before Eq. (36).

10.2 Two message-passing iterations

For iteration $a = 1, 2$,

$$Q_{ij}^a = W_Q^a E_{ij}^{a-1}, \quad (34)$$

$$K_{ij}^a = W_K^a E_{ij}^{a-1}, \quad (35)$$

$$m_{ij}^a = F^a \left(\sum_n Q_{in}^a \otimes K_{nj}^a \right) \odot G^a(E_{ij}^{a-1}), \quad (36)$$

$$h_i^a = h_i^{a-1} + U^a \left(\left[\sum_j m_{ij}^a, V_i^{a-1} \right] \right), \quad (37)$$

$$h_{ij}^a = h_{ij}^{a-1} + H^a([m_{ij}^a, E_{ij}^{a-1}]), \quad (38)$$

$$V_i^a = (v_i^0, h_i^a), \quad (39)$$

$$E_{ij}^a = (e_{ij}^0, h_{ij}^a). \quad (40)$$

The outer product in Eq. (36) is flattened before F^a . F^a, G^a, H^a, U^a are one-layer real affine maps followed by exact GELU. The two iterations have independent parameters. No division by N_e , no neighbor-count normalization, and no LayerNorm is applied. Self edges remain available to both the edge update and node aggregation. This is the literal $\sum_j$ printed in Luo–Fu Eq. (19), rather than the “all other particles” neighborhood convention of Ref. [3]. The literal physical features v_i^0, e_{ij}^0 are reinserted after every iteration.

11 Backflow, orbital transformation, and determinant

The complex backflow matrix is implemented as a real, bias-free $32 \rightarrow 4$ linear map. Its first and last two out-

puts are the real and imaginary parts of a two-component displacement,

$$\delta \mathbf{r}_i(X) = W V_i^2, \quad \tilde{\mathbf{r}}_i = \mathbf{r}_i + \delta \mathbf{r}_i. \quad (41)$$

There is no tanh, clipping, or external backflow scale. The fixed mean-field coefficients are evaluated only at this transformed coordinate,

$$\varphi_{\mathbf{k}i}(X) = \sum_{\mathbf{G}} u_{1\mathbf{k}\mathbf{G}}^{l_i} \exp[i(\mathbf{k} + \mathbf{G}) \cdot (\mathbf{r}_i + \delta \mathbf{r}_i(X))]. \quad (42)$$

Consequently, backflow does not update $u_{1\mathbf{k}\mathbf{G}}^{l_i}$. The factors $\exp[i(\mathbf{k} + \mathbf{G}) \cdot \delta \mathbf{r}_i]$ are generated only by changing the coordinate argument.

The neural orbital transformation uses the Cartesian two-component mesh momentum and the final node feature,

$$[V_i^2, k_x, k_y] \in \mathbb{R}^{34} \xrightarrow{34 \rightarrow 32} \text{GELU} \xrightarrow{32 \rightarrow 2} (J_{\mathbf{k}i}^{\text{R}}, J_{\mathbf{k}i}^{\text{I}}), \quad (43)$$

$$J_{\mathbf{k}i} = J_{\mathbf{k}i}^{\text{R}} + i J_{\mathbf{k}i}^{\text{I}}, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i} \varphi_{\mathbf{k}i}. \quad (44)$$

The direct complex output is not converted to $1 + \alpha \Delta J$. Equation (44) is implemented once: J is not included inside φ and then multiplied again. A unit test that sets $J_{\mathbf{k}i} = 1/2$ verifies $\det Q = 2^{-N_e} \det \varphi$ to numerical precision.

For the Luo–Fu full- M state,

$$\Phi_{\gamma i}(X) = \sum_{\mathbf{k}} M_{\gamma \mathbf{k}} Q_{\mathbf{k}i}(X), \quad (45)$$

$$\Psi_M(X) = \det_{\gamma i} \Phi_{\gamma i}(X), \quad M \in \mathbb{C}^{N_e \times 9}. \quad (46)$$

All graph, W , J , and complex M parameters are trainable. Only one final generalized Slater determinant is used. Because M is trainable and dense, the Cauchy–Binet expansion of $\det(MQ)$ generally contains several total-momentum sectors.

The no- M ablation removes the matrix rather than replacing it by another trainable map,

$$\Psi_{\Gamma}(X) = \det_{\alpha i} Q_{\mathbf{k}_{\alpha} i}(X), \quad \sum_{\alpha=1}^{N_e} \mathbf{k}_{\alpha} = \Gamma \pmod{\mathbf{G}}. \quad (47)$$

For $N_e = 3$, the selected fractional momenta are $\{(2,0), (2,1), (2,2)\}/3$; for $N_e = 6$ they are $\{(1,0), (1,1), (1,2), (2,0), (2,1), (2,2)\}/3$. They minimize the sum of bare band-1 energies subject to total momentum Γ . Each generalized orbital retains its own single-particle momentum; only the determinant has total momentum zero. No C_3 label or ED rotation eigenvalue is supplied.

12 Initialization and trainable parameter count

Every affine neural-network kernel uses JAX LeCunNormal, $\text{std} = 1/\sqrt{\text{fan in}}$, with zero biases. The auxiliary h^0

convention was specified above. For full- M , the real and imaginary components are independently initialized with half-scale LeCunNormal,

$$\mathbb{E}[(\Re M_{\gamma k})^2] = \mathbb{E}[(\Im M_{\gamma k})^2] = \frac{1}{2N_k}, \quad \mathbb{E}|M_{\gamma k}|^2 = \frac{1}{N_k}. \quad (48)$$

Thus every momentum minor is accessible at step zero and there is no preferred Γ subset. Ref. [1] states global LeCunNormal network initialization but does not separately specify M ; the complex extension and variance split above are therefore the present user-requested convention, not an attributed statement from the paper.

The mean-field orbital source is always band 1. However, because W and J themselves are LeCunNormal rather than forced to $W = 0, J = 1$, the step-zero many-body state is not claimed to equal the undressed bare Slater determinant pointwise. This distinction is recorded to make the initialization auditable.

Table 3: Trainable real parameter counts of the strict one-determinant implementation.

filling	N_e	full M	no M, Γ
1/3	3	78,154	78,100
2/3	6	78,208	78,100

The difference is exactly $2N_e N_k$ real parameters for the complex $N_e \times 9$ matrix M .

13 Hamiltonian and local-energy estimator

The VMC objective uses the complete continuum Hamiltonian at $\theta = 3^\circ$, screening parameter $\epsilon = 5$, and a 3×3 torus. It is not the one-band or five-band projected ED Hamiltonian. The local energy contains the kinetic term, layer-dependent moiré potential, spatially varying inter-layer tunnelling, and periodic Coulomb interaction,

$$E_{\text{loc}}(X) = \frac{(H\Psi)(X)}{\Psi(X)} = E_{\text{kin}} + E_{\text{moiré}} + E_{\text{tun}} + E_{\text{C}}. \quad (49)$$

The Coulomb sum uses reciprocal-shell cutoff 8, dielectric $\epsilon = 5$, and removes the $\mathbf{q} = 0$ neutral-background component. Tunnelling is evaluated by flipping one layer index and multiplying the exact wavefunction ratio.

Cartesian derivatives of the complete complex state use the three-point central stencil

$$\partial_a \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) - \Psi(X - h\hat{a})}{2h\Psi(X)}, \quad (50)$$

$$\partial_a^2 \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) + \Psi(X - h\hat{a}) - 2\Psi(X)}{h^2\Psi(X)}. \quad (51)$$

Here $h = 1.5 \times 10^{-3} a_M$. Periodic wrapping is applied after every coordinate displacement. A previous $h, h/2, h/4$ audit is not reused as a validation of the new checkpoints; it must be repeated after convergence.

13.1 Raw-coordinate and full-wavefunction Hamiltonian audit

The production JAX path is `src/jax_neural_bloch.py:184-248`; the independent PyTorch reference is `src/continuum_vmc.py:96-206`. Both implementations first form $\mathbf{r}_i$ from the input *raw* positions. Consequently,

$$V_M(X) = -2V \sum_{i,a} \cos(\mathbf{g}_a \cdot \mathbf{r}_i + \phi_{li}), \quad (52)$$

$$V_C(X) = \sum_{i<j} \sum_{\mathbf{q} \neq 0} v_{\mathbf{q}} \cos[\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)] \quad (53)$$

never see $\delta \mathbf{r}_i$. For tunnelling, $t(\mathbf{r}_i)$ is evaluated at that same raw coordinate, while its numerator is the complete $\Psi(X_{l_i \rightarrow 1-l_i})$ obtained by a new `logpsi` call. The kinetic ratios similarly use new complete calls,

$$R_{i\mu}^{\pm} = \frac{\Psi(X_{\mathbf{r}_i \rightarrow \mathbf{r}_i \pm \hbar \hat{\mu}})}{\Psi(X)}, \quad (54)$$

so each displaced call rebuilds the graph and recomputes every particle's $\delta \mathbf{r}_j$, $J_{\mathbf{k}j}$, the MQ contraction, and the determinant. No backflow displacement is cached by the Hamiltonian.

As a final energy-zero and convention check, a fixed-state regression sets $\delta \mathbf{r} = 0$, $J = 1$, and a fixed M selecting a known two-orbital bare Bloch Slater determinant. The four local-energy components are then compared with an independent finite-basis evaluation. This test uses 4,096 walkers, 120 burn sweeps and 10 measurements and is never used for an optimization update.

Table 4: Fixed-state Hamiltonian regression in meV for the complete many-body state. Parentheses are Monte Carlo standard errors.

term	finite-basis	Monte Carlo	difference
kinetic	39.4207	39.3467(3343)	-0.0740
moiré	-38.5168	-38.7739(5347)	-0.2571
tunnelling	-29.9435	-29.6118(2984)	+0.3317
Coulomb	-9.6769	-10.1671(3354)	-0.4902
total	-38.7164	-39.2060(3354)	-0.4896

Every component and the total agree within 1.5 quoted standard errors. The regression therefore finds no common Hamiltonian normalization, Coulomb background, tunnelling, or energy-zero mismatch at its present precision.

A targeted regression test in `tests/test_jax_neural_bloch.py` instruments the real `_encoded_graph` function. One local-energy call produces four complete graph evaluations: base X with shape $(B, N_e, 2)$, all plus and all minus configurations with shape $(2N_e B, N_e, 2)$ each, and all one-particle layer flips with shape $(N_e B, N_e, 2)$. It checks every displaced coordinate and every flipped layer against an independently constructed array. The test passes, as does the

JAX-PyTorch full-local-energy parity test to 2×10^{-7} meV absolute tolerance. No Hamiltonian input-routing error was found in this audit.

14 MCMC and natural-gradient optimization

There are 4,128 simultaneous walkers. Initial fractional coordinates are uniform on the torus and layer indices are uniform in $\{0, 1\}$. Each sweep proposes, for every walker, one random-particle Gaussian displacement with standard deviation 0.06 in fractional supercell coordinates, followed by one random-particle layer flip. Both use the Metropolis ratio $\min[1, |\Psi(X')/\Psi(X)|^2]$. The present jobs use 300 burn-in sweeps and two sweeps per update. Acceptance for coordinate and layer moves is recorded separately.

For real parameters θ_a ,

$$O_a(X) = \partial_{\theta_a} \log \Psi_{\theta}(X), \quad (55)$$

$$S_{ab} = \text{Re}[\langle O_a^* O_b \rangle - \langle O_a^* \rangle \langle O_b \rangle], \quad (56)$$

$$F_a = 2 \text{Re}[\langle (E_{\text{loc}} - \langle E_{\text{loc}} \rangle) O_a^* \rangle]. \quad (57)$$

The update is

$$(S + \lambda I) \Delta \theta = F, \quad \theta \leftarrow \theta - \eta \Delta \theta, \quad (58)$$

with ordinary, unpreconditioned CG. JAX JVP/VJP operations apply the metric matrix-free. Starting at iteration 20 and after every subsequent Krylov step, the code explicitly recomputes the authoritative residual

$$r_{\text{rel}}^{\text{true}} = \frac{\|F - (S + \lambda I) \Delta \theta\|_2}{\|F\|_2} \quad (59)$$

with an additional Fisher-vector product. The solve stops immediately when $r_{\text{rel}}^{\text{true}} < 0.02$ or at 60 CG iterations. At the cap, an update is accepted only if $r_{\text{rel}}^{\text{true}} < 0.05$ and $F^T \Delta \theta > 0$. Otherwise the parameter update is rejected and the following step uses the recovery values $\lambda = 0.03$ and $\eta = 10^{-3}$. After two consecutive stable recovery solves, the protocol returns to $\lambda = 0.01$ and $\eta = 2 \times 10^{-3}$. Every trace stores the true residual, $F^T \Delta \theta$, force and direction norms, acceptances, and whether the update was applied. This is an adaptive truncated natural gradient, not a claim that the 10^{-3} linear-system tolerance has been reached.

Table 5: Optimization and numerical settings. Quantities not specified by Ref. [1] are explicitly marked as local choices.

quantity	current value	provenance
width / message iterations	32 / 2, unshared	Ref. [1], Appendix B
orbital MLP hidden width	32, GELU	Ref. [1], Appendix B
final determinants	1	local controlled choice
optimizer / implementation	natural gradient / JAX 0.11.1	Ref. [1]
learning rate	2×10^{-3}	Ref. [1]
walkers per update	4,128	Ref. [1]
affine-kernel initialization	LeCunNormal	Ref. [1]
h^0 initialization	random, trainable	random: Ref. [1]; law/sharing: local
full- M initialization	complex LeCunNormal	user choice; paper unspecified
node aggregation	$\sum_j m_{ij}$, including self	Ref. [1], Eq. (19)
updates in present experiment	120	user-specified finite run
paper long-run statement	1,000 for clusters $> 3 \times 3$	Ref. [1]
burn / sweeps per update	300 / 2	local choice
proposal scale	0.06 fractional	local choice
SR damping	10^{-2}	local choice
CG range / true-residual rules	20–60 / 0.02 stop, 0.05 accept	user-specified
wavefunction / SR batch	516 / 516	local memory choice
local-energy batch	32	local memory choice
finite-difference step	$1.5 \times 10^{-3} a_M$	local choice
random seed	83	matched local choice
precision	JAX 64 bit	local numerical choice

All 4,128 walkers are used in the energy, force, and metric estimate at each update. The printed within-step SEM treats walker values as an instantaneous ensemble diagnostic; it is not an independent validation error. Checkpoints at initialization, every 10 updates, the latest completed update, and update 120 store both flattened parameters, optimizer recovery state, sampler RNG, and walker coordinates/layers. Interrupted jobs resume this complete state; no validation configuration ever enters an update.

15 Implementation checks completed before production

The current complete 51-test suite passes; selected architecture, parity, observable, and local-energy checks verify:

1. particle exchange changes the determinant sign and no other factor;
2. J_{ki} is applied exactly once;
3. fixed Bloch coefficients are absent from trainable parameters;
4. shifted-coordinate Bloch evaluation agrees with an explicit plane-wave calculation;
5. PyTorch and JAX $\log \Psi$ agree at identical parameters;
6. complete JAX and PyTorch local energies agree;
7. no- M is invariant under both primitive translations and contains no trainable M array;
8. the exact C_3 projector implementation remains available but is not used in the present no- M experiment;
9. matrix-free force and Fisher-vector products agree with an explicit score-Jacobian calculation;
10. checkpoint export between JAX and PyTorch is lossless;

11. full- M is reproducibly random, complex, and not one-hot;
12. h_i^0, h_{ij}^0 are nonzero trainable parameters; and
13. a one-particle graph receives its nonzero $j = i$ message in Eq. (19);
14. the 1-RDM uses full replaced-configuration ratios and a fixed bare projector, and its bare-Slater full-basis trace is N_e ; and
15. kinetic displacements and tunnelling flips rebuild the complete JAX graph, while diagonal Hamiltonian factors use raw coordinates.

A one-update end-to-end smoke test also passed MCMC, local-energy evaluation, truncated SR, residual recording, parameter update, and checkpoint writing. These checks validate implementation consistency, not physical convergence.

16 Controlled four-run 120-update experiment

Four jobs were trained from independent random walker ensembles with the same network seed and numerical protocol. They differ only in filling and in the final momentum map. Full M has a dense trainable complex momentum matrix and therefore mixes many-body momentum sectors; no M removes this matrix and uses the fixed orbital set whose total momentum is exactly Γ . Table 6 lists the reproducible result directories. Every directory contains the full trace, checkpoints at intervals of ten attempted updates, final parameters, persistent walkers and sampler RNG state.

Table 6: Authoritative 120-update calculations used in this report.

ν	ansatz	K_{tot}	result-directory suffix
1/3	full M	mixed	adaptive_cg120_20260821_full_m_nu1of3
1/3	no M	Γ	adaptive_cg120_20260821_no_m_gamma_nu1of3
2/3	full M	mixed	adaptive_cg120_20260821_full_m_nu2of3
2/3	no M	Γ	adaptive_cg120_20260821_no_m_gamma_nu2of3

16.1 Training-chain diagnostics

Table 7 summarizes the final ten attempted updates. The mean and RMS describe the persistent training chain and are not independent energy estimates. A rejected update contributes its sampled local energy to the diagnostic trace but leaves the parameters unchanged. The two $\nu = 1/3$ jobs never activated the rejection rule. At $\nu = 2/3$, both jobs rejected 19 of 120 attempted updates; the final full- M attempt was rejected because $r_{\text{rel}}^{\text{true}} = 0.0512$ exceeded the 0.05 cap. The saved step-120 checkpoint therefore contains the parameters after the last accepted update, not the rejected direction.

Table 7: Final-ten training diagnostics. Energies are meV per particle; RMS is training-chain scatter.

ν	ansatz	$\langle E/N_e \rangle$	min. E/N_e	RMS	$\langle r_{\text{rel}} \rangle$	$\langle N_{\text{CG}} \rangle$	reject ₁₂₀
1/3	full M	-35.9952	-36.0866	0.0434	0.0183	48.0	0
1/3	no M, Γ	-36.7708	-36.9023	0.0592	0.0201	56.7	0
2/3	full M	-52.0575	-52.1563	0.0817	0.0371	54.5	19
2/3	no M, Γ	-52.4147	-52.4842	0.0578	0.0397	54.9	19

The final-ten mean position/layer acceptances are 0.681/0.592 and 0.669/0.618 at $\nu = 1/3$, and 0.651/0.604 and 0.660/0.632 at $\nu = 2/3$ (full M , no M order). Local-energy variances remain sizeable: the corresponding final-ten means are 118.0, 104.5, 701.0 and 718.3 meV². Thus a smooth energy trace does not establish a zero-variance eigenstate.

16.2 Independent validation and checkpoint selection

Training walkers are never used to select the reported final energy. After all four step-120 checkpoints were frozen, each was tested with 1,024 new random walkers, 200 burn-in sweeps and ten measurement sweeps. A local energy was evaluated after every measurement sweep, giving 10,240 measurements per state. Within each filling, the lower re-equilibrated low-cost energy selects the one state that receives a second validation with 4,128 new walkers, 300 burn-in sweeps and 20 measured sweeps (82,560 local energies). The selection is therefore independent of the minimum training-chain point.

Table 8: Independent validation energies in meV per particle. Parentheses are eight-block, sweep-major standard errors. ΔE_{5b} uses the matching five-band ED energy.

ν	ansatz	low-cost E/N_e	precise E/N_e	variance	ΔE_{5b}
1/3	full M	-36.14253(4822)	—	113.90	+1.25071
1/3	no M, Γ	-36.79290(2589)	-36.78317(1019)	120.65	+0.61006
2/3	full M	-52.01665(4485)	—	730.94	+0.70889
2/3	no M, Γ	-52.39480(7574)	-52.39973(1673)	700.60	+0.32581

Both selection stages favor no M, Γ . However, the precise results remain above five-band ED by 0.6101 and 0.3258 meV per particle. Therefore the present calculation does not demonstrate a neural-Bloch variational energy advantage. It also does not prove that no M is asymptotically more expressive: the comparison contains one parameter initialization, only 120 attempted updates and many rejected $\nu = 2/3$ directions.

17 Post-training one- and two-body observables

Figure 7 displays both step-120 variants: full M from the independent 1,024-walker low-cost validation ensembles and no M, Γ from the independently re-equilibrated 4,128-walker precise ensembles. The quantitative 1-RDM table and the numerical estimates below use the selected

precise no- M states. No persistent training configuration enters any displayed observable.

17.1 Full-ratio 1-RDM and fixed bare-band projector

For $a = (n, \mathbf{k})$ and $b = (m, \mathbf{q})$, the estimator is

$$\begin{aligned} \rho_{ab} &= \langle c_b^\dagger c_a \rangle \\ &= 2N_e \mathbb{E}_{X \sim |\Psi|^2, i, x'} \left[(\phi_a^0(x_i))^* \phi_b^0(x') \left(\frac{\Psi(X_{i \rightarrow x'})}{\Psi(X)} \right)^* \right]. \end{aligned} \quad (60)$$

The factor two accounts for uniform auxiliary sampling of the two layers. In every ratio, replacing one particle triggers a complete new ansatz call: the code rebuilds the graph and recomputes all particles' $\delta \mathbf{r}_j$, $J_{\mathbf{k}j}$, the optional MQ contraction and the determinant. The projector is instead the fixed bare function

$$\phi_{n\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{n\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{r}}; \quad (61)$$

no backflow phase is multiplied into it. Thus neural dressing appears once, only through the complete wavefunction ratio.

All 4,128 validation configurations and 16 auxiliary points per configuration are used to estimate the 45×45 five-band-projected matrix. The band populations are

$$W_n = \frac{1}{N_e} \sum_{\mathbf{k}} \rho_{n\mathbf{k}, n\mathbf{k}}. \quad (62)$$

Table 9: Five-band-projected 1-RDM results for the selected states. Band weights are percentages; errors are eight-block estimates.

ν	trace	W_1	W_2	W_3	W_4	W_5	min eig.
1/3	2.986(95)	96.63(94)	3.70(1.08)	-0.40(1.02)	0.27(1.06)	-0.65(71)	-0.193
2/3	6.266(176)	89.12(86)	8.38(1.42)	3.77(65)	1.87(77)	1.29(1.34)	-0.172

The projected traces are statistically compatible with N_e at the stated precision, but the matrix is only Hermitized, not projected onto the positive semidefinite cone. Negative eigenvalues and the small negative $\nu = 1/3$ remote-band weights expose remaining auxiliary-coordinate noise. Consequently, no natural-orbital eigenspace is interpreted as stable. The more robust diagonal populations indicate approximately 3.4% and 10.9% weight outside bare band 1, respectively.

The focused bare-Slater regression sets $\delta \mathbf{r} = 0$, $J = 1$, fixes a complex 2×9 matrix M with orthonormal rows, and samples the resulting single determinant. In a complete nine-orbital test basis the exact matrix is $\rho = M^\top M^*$, with trace two. With 4,096 walkers, 100 burn sweeps and 32 auxiliary draws, the measured trace is 2.01047 ± 0.01459 , the Frobenius error is 0.05272, the Hermiticity residual is zero, and the minimum sampled eigenvalue is -0.02072 . This unit test verifies occupation, Hermiticity, PSD within sampling noise and full-basis trace normalization.

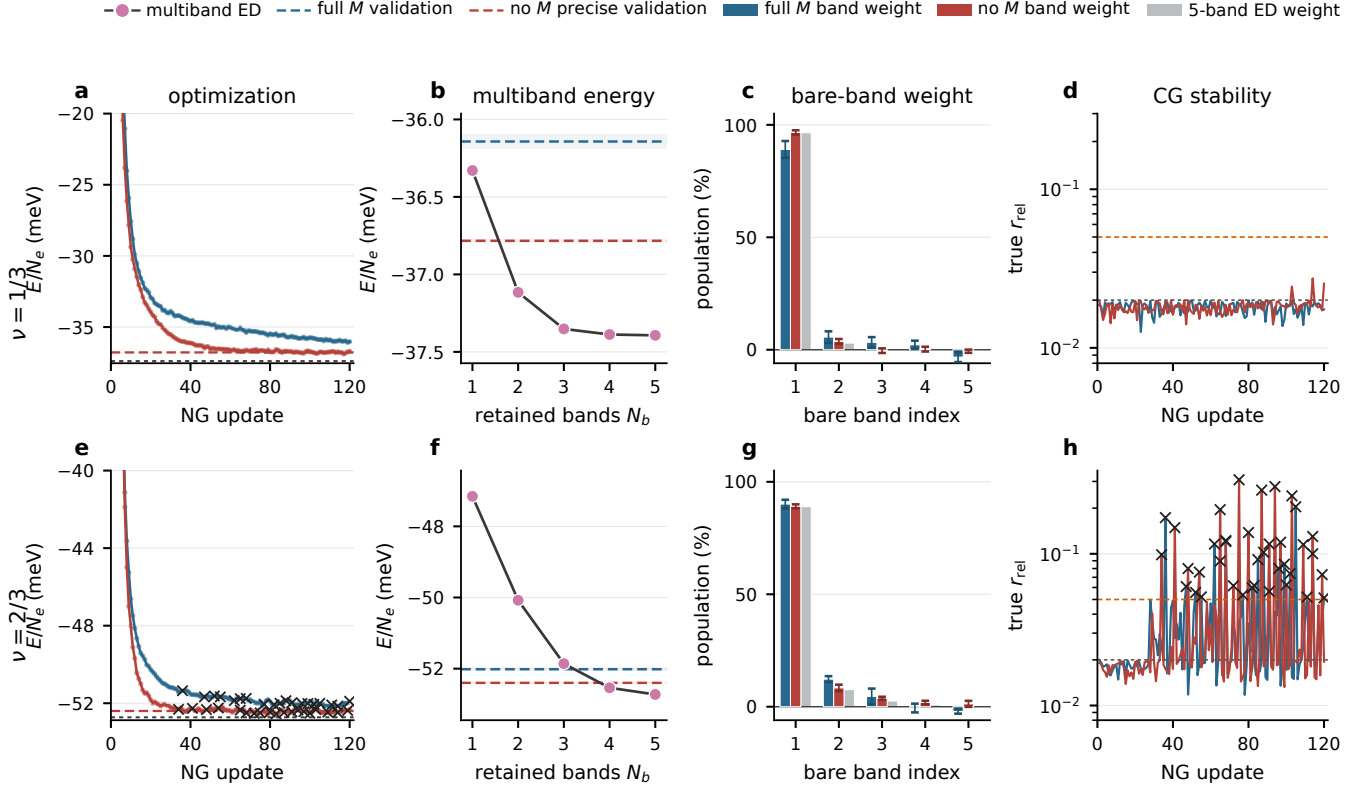


Figure 6: Controlled neural-Bloch optimization, multiband energy, band mixing and CG stability. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,e) Persistent-walker training energies; black crosses mark rejected updates, the gray dotted line is five-band ED, and the red dashed line (with a faint uncertainty band) is the independently re-equilibrated precise no- M validation energy. The upper limits remain -20 and -40 meV per particle. (b,f) One- through five-band ED energies, together with the independent low-cost full- M validation (blue dashed) and precise no- M validation (red dashed). (c,g) Bare-band populations from the complete-wavefunction-ratio 1-RDM: step-120 full M from its 1,024-walker independent validation ensemble, precise no M from 4,128 fresh walkers, and five-band ED. Error bars are block estimates. (d,h) Explicitly recomputed true CG residuals; horizontal lines mark the 0.02 early-stop and 0.05 cap-acceptance thresholds. Training-chain minima are not used as final energies.

17.2 Full structure factor, connected part, and density order

The structure factor is evaluated directly as a two-body observable, not reconstructed from the 1-RDM:

$$S_{\text{full}}(\mathbf{q}) = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle}{N_e}, \quad (63)$$

$$S_c(\mathbf{q}) = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle - |\langle \rho_{\mathbf{q}} \rangle|^2}{N_e}, \quad (64)$$

$$m_{\mathbf{q}} = \frac{|\langle \rho_{\mathbf{q}} \rangle|}{N_e}. \quad (65)$$

The ED comparison uses S_{full} with the same normalization. Its largest nonzero-momentum values are 1.5060 ± 0.0118 at $\nu = 1/3$ and 0.6760 ± 0.0106 at $\nu = 2/3$, compared with five-band ED values 1.4381 and 0.6197. The corresponding maximum nonzero- q one-body amplitudes are only 0.0073 and 0.0039. Therefore the Γ states have negligible static translation-breaking expectation at the sampled reciprocal vectors; their large S_{full} peaks are density fluctuations. The real-space panels in Fig. 7 include

the intra-moiré-cell Bloch density and should not be identified with a nonzero CDW order parameter by visual contrast alone.

18 Conclusion

The implemented ansatz now follows the printed Luo-Fu Appendix-B structure: fixed mean-field band-1 plane-wave coefficients, periodic node and edge features, random trainable auxiliary features, two unshared width-32 message iterations with the literal self-inclusive $\sum_j m_{ij}$, coordinate backflow, direct complex $J_{\mathbf{k}i}$ applied exactly once, and one final generalized Slater determinant. The two tested QNS differ only in their final momentum map: a trainable complex matrix M versus a fixed Γ orbital set.

The fixed-state componentwise Hamiltonian regression finds no energy-zero, Coulomb-background or input-routing mismatch. The controlled adaptive-CG experiment also prevents poorly solved directions from silently changing the parameters and records all rejected updates. Nevertheless, independent validation gives $E/N_e =$

$-36.78317(1019)$ and $-52.39973(1673)$ meV for the selected $\nu = 1/3$ and $2/3$ no- M states. These remain 0.6101 and 0.3258 meV per particle above five-band ED. The present neural-Bloch calculation therefore has no established variational energy advantage.

The validated 1-RDM indicates weak remote-band population at $\nu = 1/3$ and stronger mixing at $\nu = 2/3$, while the full structure factors retain the CDW/FCI benchmark contrast. However, noisy non-PSD 1-RDM eigenvalues, one initialization, finite optimization and substantial rejected-update rates at $\nu = 2/3$ preclude a converged natural-orbital or topological interpretation. The next controlled step is several independent seeds with a better conditioned SR model and independently equilibrated force estimates; extending a single two-sweep persistent chain is not sufficient evidence.

References

- [1] D. Luo, T. Zaklama, and L. Fu, "Solving fractional electron states in twisted MoTe_2 with deep neural network," arXiv:2503.13585v3, [doi:10.48550/arXiv.2503.13585](https://doi.org/10.48550/arXiv.2503.13585).
- [2] X. Li, Y. Chen, B. Li, H. Chen, F. Wu, J. Chen, and W. Ren, "Deep Learning Sheds Light on Integer and Fractional Topological Insulators," arXiv:2503.11756v1, [doi:10.48550/arXiv.2503.11756](https://doi.org/10.48550/arXiv.2503.11756).
- [3] G. Pescia, J. Nys, J. Kim, A. Lovato, and G. Carleo, *Message-Passing Neural Quantum States for the Homogeneous Electron Gas*, arXiv:2305.07240v3, [doi:10.48550/arXiv.2305.07240](https://doi.org/10.48550/arXiv.2305.07240).
- [4] D. Luo and B. K. Clark, "Backflow transformations via neural networks for quantum many-body wave functions," *Phys. Rev. Lett.* **122**, 226401 (2019), [doi:10.1103/PhysRevLett.122.226401](https://doi.org/10.1103/PhysRevLett.122.226401).
- [5] A. P. Reddy *et al.*, "Fractional quantum anomalous Hall states in twisted bilayer MoTe_2 and WSe_2 ," *Phys. Rev. B* **108**, 085117 (2023), [doi:10.1103/PhysRevB.108.085117](https://doi.org/10.1103/PhysRevB.108.085117).
- [6] Z. Lin *et al.*, "Fractional Chern insulator states in an isolated flat band of zero Chern number," *Newton* **2**, 100339 (2026), [doi:10.1016/j.newton.2025.100339](https://doi.org/10.1016/j.newton.2025.100339).
- [7] L. Zhang, G. Duan, and D. Luo, "WF-Bench: A benchmark for neural network wavefunction expressivity and scaling laws," arXiv:2605.29683 (2026), [doi:10.48550/arXiv.2605.29683](https://doi.org/10.48550/arXiv.2605.29683).
- [8] M. Yamazaki *et al.*, "Physics-inspired transformer quantum states via latent imaginary-time evolution," *Phys. Rev. Research* **8**, 033032 (2026), [doi:10.1103/bjxb-8tsk](https://doi.org/10.1103/bjxb-8tsk).
- [9] R. Rende, L. Viteritti, and A. Georges, "Superconductivity in the t - t' Hubbard model from symmetry-preserving neural-network quantum states," arXiv:2608.12465 (2026), [doi:10.48550/arXiv.2608.12465](https://doi.org/10.48550/arXiv.2608.12465).

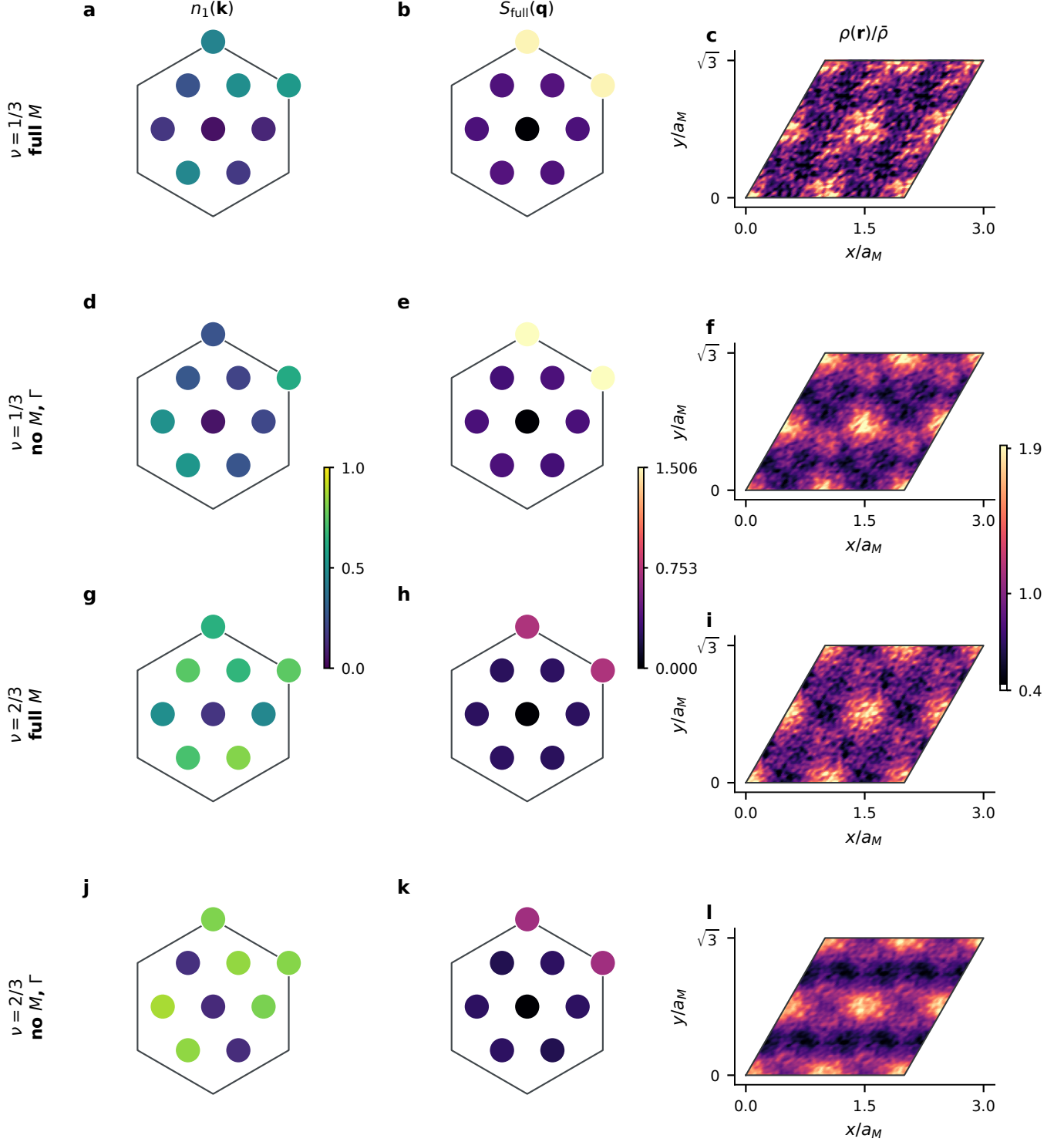


Figure 7: Full- M and no- M observables from independent validation ensembles. Rows are, in order, $\nu = 1/3$ full M , $\nu = 1/3$ no M , $\nu = 2/3$ full M , and $\nu = 2/3$ no M . (a,d,g,j) Bare-band-1 momentum occupation from the five-band-projected complete-ratio 1-RDM. (b,e,h,k) Full static structure factor with the ED definition on the complete hexagonal BZ. (c,f,i,l) Folded charge density over the complete two-by-two moiré-cell parallelogram. Color normalization is shared within each column. Full- M panels use 1,024 fresh validation walkers and no- M panels use the 4,128-walker precise ensembles; no persistent training walker enters the estimator.